

In my project, I tested a 3x3 grid and a 4x4 grid on 2 uniformed search algorithms, uniformed cost and BFS, and on 3 informed search algorithms, A* with Manhattan-Distance heuristics, Greedy BFS search, and best first search. Overall, the run times, nodes expanded, and solution length were much longer on a 4x4 grid then the 3x3. This was an expected result since 4x4 would obviously take more time, however, what surprised me the most was how much longer the run times were. For algorithms like BFS and DFS, the run times were almost too much for my computer to run and in many cases took up all the RAM on my PC causing me to have to stop the program. I believe that this is because this algorithm searches through each row/column which in a 4x4 grid there are many more.

Each of my random problems were randomized which a vary amount of steps. As the number of randomized steps increased, so did the run time and the lengh of the solution. For example, my 4x4 grid using uniform cost search had a length of solution for 5 randomized steps of 5,5, and 5. However, with the 3 grids with 40 steps of randomization, the lengths of the solutions were 18, 16, and 20. Based off this observation, I originally thought that the more steps of randomization used the longer it will take search algorithms to complete, however, I believe that the more steps of randomization will increase the run time but only to a certain point as once a table had been thoroughly randimized, the run time will remain the same.

I observed that BFS and Unformed cost search had a greater number of nodes explored and longer lengths of solution (to the point where all the RAM in my PC was used). Both of these algorithms are unguided and just search till they find a final state. For the informed search algorithms, they use manhattan distance or out of place heuristics to guide them to a solution.

Overall, informed search algorithms have much shorter run times, length of solutions, and nodes expanded. The size of the grid greatly affects this features just as well as the search algorithms. As the grid size increases the run time expands exponentially which is why unformed search algorithms should not be used.